Chaeun Park

chaeunnara@snu.ac.kr Seoul, South Korea chaephymath

Education

Seoul National University, Physics and Mathematical Sciences (Double Major) — Seoul, South Korea
- College of Natural Sciences — 2022 – Aug 2027

Research Experience

Computational Study of Pseudomodular Fuchsian Groups, Research Internship — SNU, Seoul, South Korea
Mar 2026 – Aug 2026

Advised by Prof. Gye-Seon Lee. Studied the construction of pseudomodular Fuchsian groups through Long–Reid's "Pseudomodular Surfaces," focusing on rational cusp sets, arithmeticity, and commensurability. Developed Python/SymPy calculations for trace-based arithmeticity tests, rational fixed-point searches, and killer-interval coverings. Reproduced computational certificates for $\Delta(2/5, 4)$ and $\Delta(3/7, 4)$, and performed an appendix-guided verification for $\Delta(3/11, 4)$. Documented the methods and results in an undergraduate research internship report.

- Studied constructions of pseudomodular Fuchsian groups and the relationship between rational cusp sets, arithmeticity, and commensurability.
- Developed Python/SymPy computations to reproduce trace-based arithmeticity tests, detect rational fixed points of hyperbolic elements, and construct killer-interval coverings.
- Reproduced computational certificates for the pseudomodularity of $\Delta(2/5, 4)$ and $\Delta(3/7, 4)$, with an appendix-guided verification for $\Delta(3/11, 4)$; documented the work in an internship report.

Local Well-Posedness of Incompressible Euler Equations on Singular Domains, Research Internship — SNU, Seoul, South Korea
Mar 2025 – Aug 2025

At SNU, advised by Prof. Injee Jeong (now at KIAS). Investigated local well-posedness of the incompressible Euler equations in Sobolev spaces via velocity–pressure methods. Studied energy estimates and pressure regularity in the whole space, half-space, and domains with corners, including boundary compatibility and elliptic regularity issues. Prepared a research report and presented the work.

- Investigated local well-posedness of incompressible Euler equations via velocity–pressure methods and Sobolev estimates.
- Studied energy and pressure estimates in whole-space, half-space, and corner domains, including boundary-regularity issues.
- Prepared a research report and gave a presentation on the problem.

Teaching & Mentoring

Topology Mentoring, Mentor — Seoul, South Korea
Apr 2026 – Aug 2026

Private, in-person undergraduate mentoring based on Chapters 2–8 of Munkres' Topology.

Algebra Mentoring, Mentor — Seoul, South Korea
Jan 2024 – Feb 2024

Private, in-person mentoring for two students, covering selected topics from Chapters 1–14 of Dummit and Foote's Abstract Algebra.

Academic Study Groups

Advanced Quantum Field Theory, Lead Organizer and Presenter — Seoul National University, Seoul, South Korea
Jan 2026 – present

Co-leading an advanced, year-long study of quantum field theory with a small group of core presenters. Topics are organized primarily around Ramond, with selected material from Schwartz, Srednicki, and other references.

Introductory Quantum Field Theory, Presenter — Seoul National University, Seoul, South Korea
Participated in a rotating seminar-style introduction to quantum field theory based on Ramond's Field Theory: A Modern Primer, with Professor Seok Kim invited as faculty mentor. — Jan 2025 – Feb 2025

Compact Lie Group Representation Theory, Presenter — Seoul National University, Seoul, South Korea
Participated in a rotating seminar based on Representations of Compact Lie Groups. — Sept 2024 – Dec 2024

Clubs

SQRT (SNU Quantum Research Team), Member — Seoul National University, Seoul, South Korea
Participated in club study sessions on quantum computing, including fundamentals of quantum information theory and superconducting quantum circuits. Recognized as an SQRT alumnus (certificate issued February 26, 2025). — Sept 2022 – Jan 2025

Convergence — Mathematics Academic Club, Member, SEGL Project — SNU, Seoul, South Korea
Participated in the SEGL (Seoul National University Experimental Geometry Lab) project within the Convergence mathematics club, advised by Prof. Gye-Seon Lee. Studied the theory of discrete reflection groups in hyperbolic geometry and primarily contributed theoretical analysis and hand calculations to a collaborative Python visualization of limit sets, drawing on Hohlweg–Labbé–Ripoll's work on infinite Coxeter groups. — Jan 2023 – Jan 2023

- Studied the theory of discrete hyperbolic reflection groups and performed hand calculations for a collaborative Python-based limit-set visualization.
- Participated in the SEGL project within Convergence; faculty adviser: Prof. Gye-Seon Lee.

Presentations & Posters

The Power of Symmetry: From Rotations to Particles, Forces, and Gravity — CERN, Geneva, Switzerland, July 2026
Joint presentation with Drim Choi and Seokwon Lee at the CERN–CKC Summer Student Programme 2026. Presented Part III, "Massless Spin-2 and Einstein Gravity," on how gauge invariance and the self-consistent coupling of a massless spin-2 field lead from Fierz–Pauli theory to Einstein gravity.

Parabolic Krylov–Safonov Harnack Inequality — Seoul National University, Seoul, South Korea, June 2026
Final presentation for Prof. Sun-Sig Byun's special-topics course on regularity methods for nonlinear elliptic and parabolic equations. Based on Beomjun Choi's notes, covered parabolic ABP estimates, measure and propagation lemmas, Vitali covering, level-set decay, weak and full Harnack inequalities, and Hölder regularity.

Euler Equations on 2D Singular Domains — Seoul National University, Seoul, South Korea, Dec 2025
Final presentation for Prof. Sun-Sig Byun's special-topics course on regularity theory for partial differential equations. Using Zonglin Han's dissertation, reviewed Yudovich well-posedness on nonsmooth and corner domains, log-Lipschitz velocity estimates, Lagrangian flows and the Osgood criterion.

PIGE Analysis of $^{12}C(p,\gamma)^{13}N$ and $^{27}Al(p,p'\gamma)^{27}Al$, $^{27}Al(p,\alpha\gamma)^{24}Mg$ Reactions using Pelletron Accelerator — SNU, Seoul, South Korea, Dec 2024
Poster presentation with Haeun Chung at the 2024 Undergraduate Research Presentation, based on experimental nuclear physics research conducted during Nishina School 2024.

Honors and Awards

Excellence Award, 2024 Undergraduate Research Presentation — Seoul National University, Seoul, South Korea, Dec 2024
Awarded by the Department of Physics and Astronomy for the poster "PIGE Analysis of $^{12}C(p,\gamma)^{13}N$ and $^{27}Al(p,p'\gamma)^{27}Al$, $^{27}Al(p,\alpha\gamma)^{24}Mg$ Reactions using Pelletron Accelerator," based on research conducted during Nishina School 2024.

Mathematics Schools & Workshops

The 6th International Undergraduate Mathematics Summer School — Seoul National University, Seoul, South Korea
Participant — 2026-07-27 to 2026-08-07

The 12th Korea PDE Winter School — POSTECH, Pohang, South Korea
Participant — 2026-01-19 to 2026-01-23

The 3rd International Undergraduate Mathematics Summer School — Seoul National University, Seoul, South Korea
Participant — 2023-08-07 to 2023-08-18

10-10 Summer School on Mathematics of Deep Learning and AI — Seoul National University, Seoul, South Korea
Participant — 2023-08-07 to 2023-08-08

Physics Schools & Workshops

CERN-Korean Summer Student Program — CERN, Geneva, Switzerland
Participant — 2026-06-29 to 2026-07-11

SNU-APCTP Winter School on Fundamental Physics — Seoul National University, Seoul, South Korea
Participant — 2026-02-02 to 2026-02-06

Nishina School 2024 — RIKEN, Japan
Participant — 2024-07-25 to 2024-08-02

Research Interests

Nonlinear PDEs and Fluids
Existence, regularity, and nonlinear dynamics of fluid equations, particularly the Euler equations, with attention to the effects of boundaries and singular geometry.

Hyperbolic Geometry and Geometric Group Theory
Discrete group actions on hyperbolic spaces, especially Fuchsian groups, and how group-theoretic and representation-theoretic properties are reflected in geometry, topology, and dynamics.

Quantum Field Theory and Gravitation
The relation between quantum field theory and spacetime geometry, from conserved quantities such as Komar integrals and ADM mass to holographic duality in AdS/CFT. I am also interested in how black-hole physics connects quantum chaos, random matrix theory, and SYK models.

Languages

Korean
Native

English
Academic working proficiency